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Flow Control and Data link layer using Sliding window

Aim: To write a program to implement flow control at Data link layer using Sliding window protocol and to Simulate the flow of frames from one node to another.

Algorithm:

1) Start the program.
2) Read the total number of frames to be Sent (N), the Sender window Size (W) and a frame number to Simulate as lost during transmission.
3) Initialize base = 0 (Sequence no. of the oldest unacknowledged frame) and next Seq = 0 (Sequence number of the next frame to be sent.
4) Repeat While base < N :
5) Repeat Step 4 until all N frames have been transmit
6) Stop the program.

Program:

```
def Simulate (total_frames, window_size=1, lost[redacted]_frame=1)
base = 0
next_seq = 0
While base < total_frames:
window_end = min (base + window_size - 1, total_frame - 1)
print ("\n . . . . . ")
Print ("[ Sender window: [Frame {base}] - Frame {window_end}]"
```

Output:

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (...
Enter total number of frames to Send : 8
Enter Sender's Window Size : 2
Enter frame number to Simulate as LOST
(0 to 7, enter -1 for no loss) : 5
```

Sender Window : [Frame 0 - Frame 1]

Node A --> Sending Frame 0

Node A --> Sending Frame 1

Node B --> Received Frame 0, Sending ACK 0

Node B --> Received Frame 1, Sending ACK 1

Node A --> ACK 1 received. Sliding window forward.

Sender Window : [Frame 2 - Frame 3]

Node A --> Sending Frame 2

Node A --> Sending Frame 3

Node B --> Received Frame 2, Sending ACK 2

Node B --> Received Frame 3, Sending ACK 3

Node A --> ACK 3 received. Sending window forward.

Sender Window : [Frame 4 - Frame 5]

Node A --> Sending Frame 4

Node A --> Sending Frame 5

(Frame 5 Lost in transit!)

Node B --> Frame 5 missing. Discarding out-of-order frames after it

Node A --> Timeout! Resending window Starting from Frame 5.

Sender Window : [Frame 4 - Frame 5]

Node A --> Sending Frame 5

Node B --> Received Frame 4, Sending ACK 4

Node B --> Received Frame 5, Sending ACK 5

Node A --> ACK 5 received. Sliding Window forward.

Sender Window : [Frames 6 - Frame 7]

Node A --> Sending Frame 6

Node A --> Sending Frame 7

Node B --> Received Frame 6, Sending ACK 6

Node B --> Received Frame 7, Sending ACK 7

Node A --> ACK 7 received. Sliding Window forward.

All 8 frames Successfully transmitted from Node A to Node B.

```
While next_seq < base + window_size and next_seq < total_frames:
    print (f'Node A --> Sending frame {next_seq}')
    if Next_seq == lost_frame:
        print(f"    (Frame {next_seq} LOST in transit!)")
        # next_seq += 1
    if lost_frame is not None and base <= lost_frame < next_seq:
        print (f" Node B --> Frame {lost_frame} missing. Discarding out-of-order
            frames after it.")
        print(f" Node A --> Timeout ! Resending Window starting from
                Frame {lost_frame}.")
        next_seq = lost_frame
        lost_frame = None
        continue
    for i in range (base, next_seq):
        print (f" Node B --> Recieved Frame {i}. Sending ACK{i
    print (f" Node A --> ACK {next_seq-1} received. Sliding Window
                                                forward.")
    base = next_seq
    print ("\n. . . . . .")
print (f"All {total_frames} frames Successfully transmitted from
                                    Node A to Node B.")

def main():
    print ("===Sliding Window Protocol - Flow Control Simulation
                                    (Go-Back-N) ===\n")
    total_frames = int(input("Enter total number of frames to
                                                Send:"))
```

```
window_size = int(input("Enter Sender's Window Size : "))
lost_frame = int(input("Enter frames number to Simulate as LOST
        \n (0 to {total_frames - 1}, enter -1 for no loss): "))
if lost_frame == -1:
    lost_frame = None
    Simulate(total_frame, window_size, lost_frame)
if __name__ == "__main__":
main()
```

Result:

The program to implement flow control at the Data Layer using the Sliding Window protocol was Successfully written in python, executed and verified by Simulating the flow of frames between two nodes (Node A as Sender and Node B receiver).